CIS437 - Cloud Computing

– Connecting Cloud Networks –

Shared VPC, VPC Network Peering, VPN, and Interconnect

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Based on the original version by Professor Erik Fredericks

Last Class

- Introduction to Networking in the Cloud
- VPCs and Subnets
- IP Addresses and CIDR
- Google Cloud Network Architecture
- Routes and Firewall Rules

Agenda

- Network Tags and Firewall Layers
- Why Multiple VPC Networks?
- Shared VPC and VPC Network Peering
- Hybrid Cloud Networking
- Cloud VPN
- Dedicated and Partner Interconnect
- Comparing Connectivity Options
- Practice

Network Tags and Firewall Rules

Setting	Example
VM name	web-vm
VM network tag	web-server
Firewall rule network	app-vpc
Direction/action	Ingress / Allow
Source	0.0.0.0/0
Protocol/port	TCP 80
Target tag	web-server

Key point: A VM name is not a network tag. Firewall rules can target VMs by network tag.

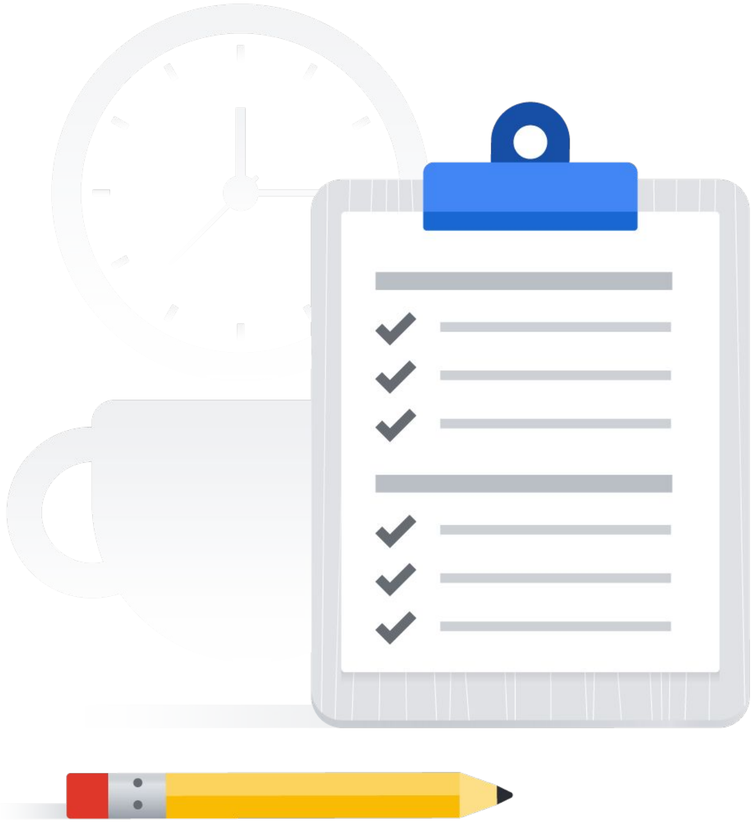
VPC Firewall vs. VM Firewall

Browser → VPC firewall → VM guest firewall (e.g., Ubuntu UFW), if enabled → web server

- Does the VPC firewall permit the connection?
- Does the guest firewall permit it, if active?
- Is the web server running and listening on the expected port?

Lab: VPC Networks — Controlling Access

Create NGINX web servers, control external HTTP access to the web servers using tagged firewall rules, and explore IAM roles and service accounts.
The lab can be found here.



Lab objectives

- **Create an NGINX web server.**
- **Create tagged firewall rules.**
- Create a service account with IAM roles.
- Explore permissions for the Network Admin and Security Admin roles.

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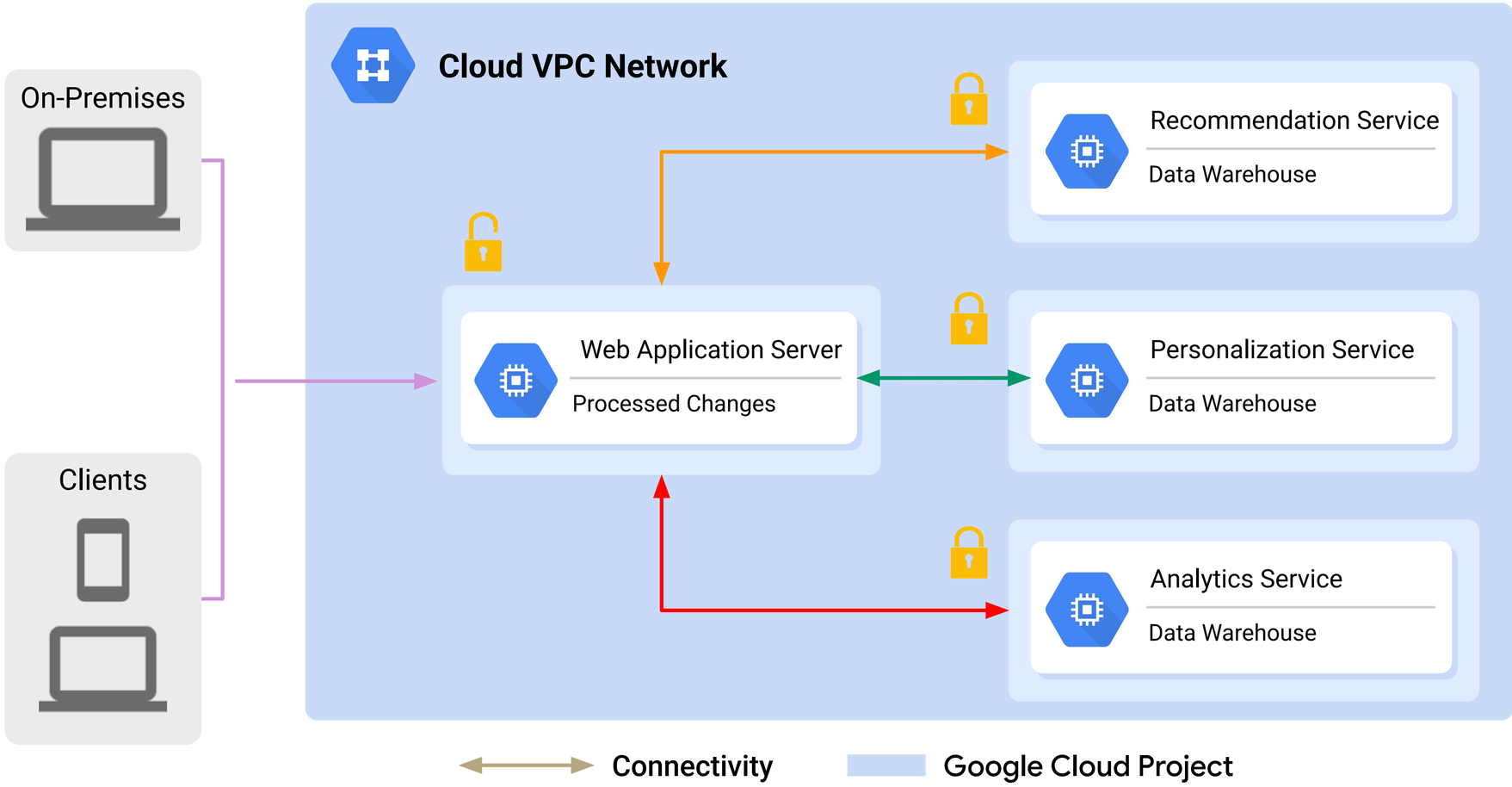
Why Multiple VPC Networks?

If one VPC is already global, why would an organization need more than one VPC?



Shared VPC: One Network, Multiple Projects.

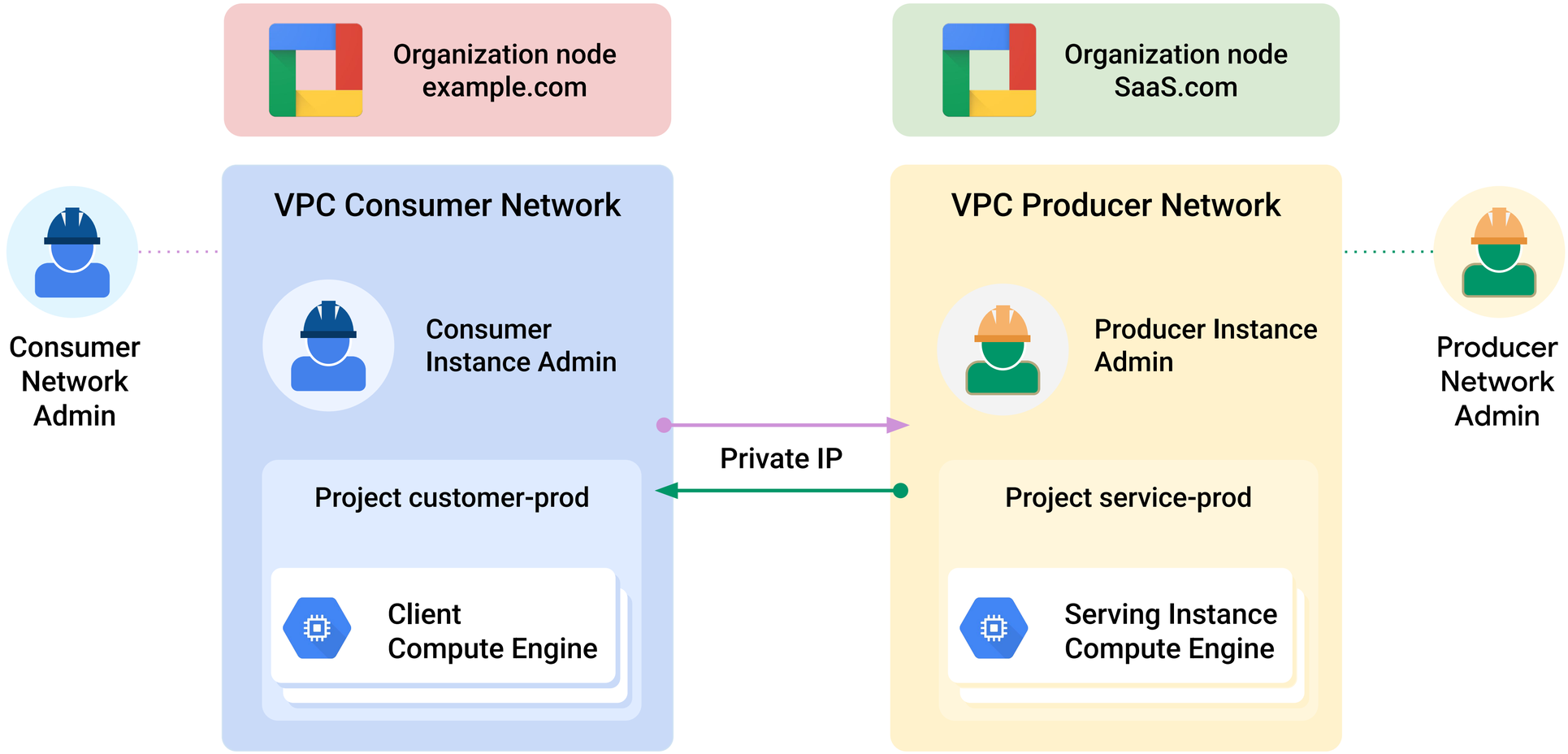
- Shared VPC = one centrally managed VPC shared by multiple projects
- Host project: owns the Shared VPC
- Service projects: use subnets from that VPC



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VPC Network Peering: Connect Separate Networks.

- Peering connects separate VPC networks using internal IP addresses.
- Each VPC keeps its own firewall rules and administration.



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Considerations for VPC Network Peering

- Peered VPCs remain administratively separate.
- Both sides must configure the peering relationship.
- Subnet CIDR ranges **must not** overlap.
- Works with Compute Engine, Google Kubernetes Engine, and App Engine flexible environments.

- *Peering connects networks, but does not merge their administration.*
- *Plan non-overlapping CIDR ranges before peering.*



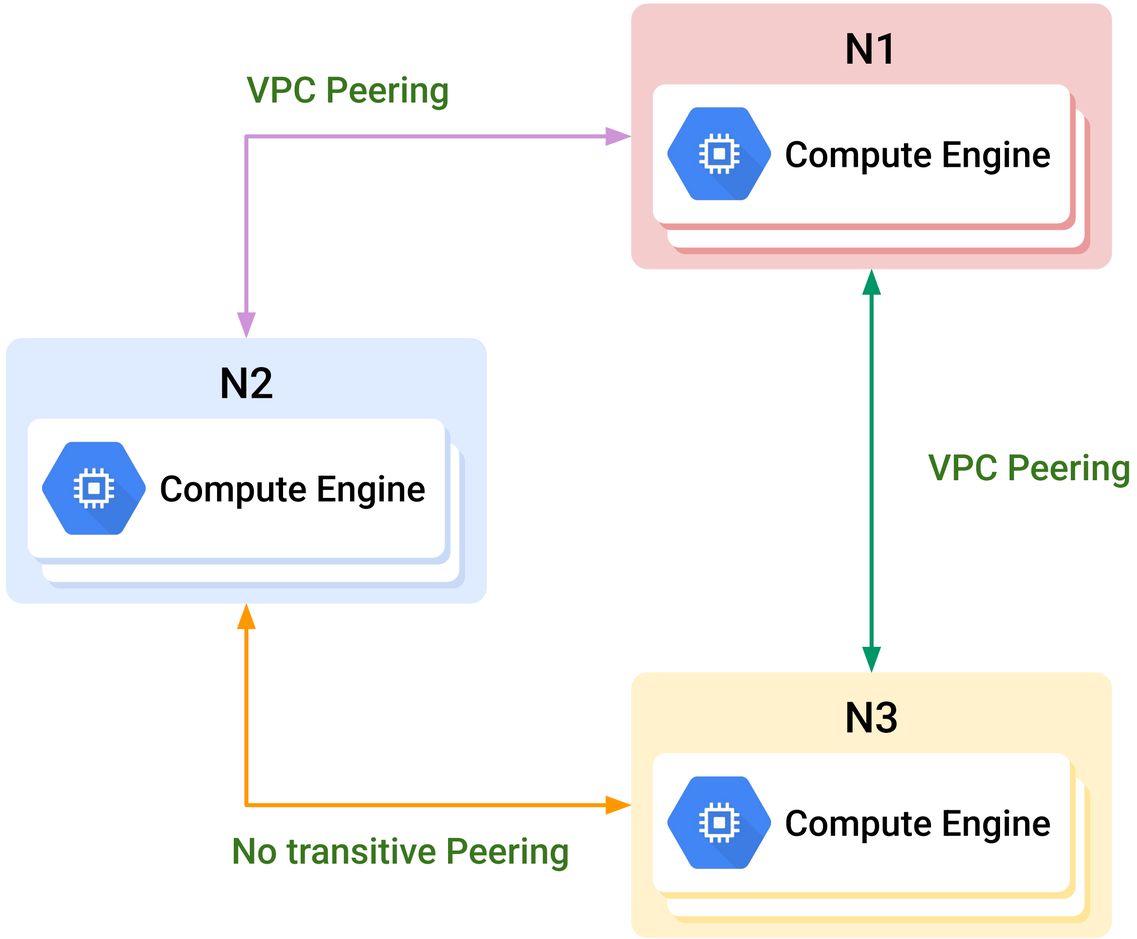


If Network 1 peers with Network 2, and Network 2 peers with Network 3, can Network 1 automatically reach Network 3?

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VPC Peering Is Not Transitive

N2 ↔ N1 ↔ N3 does NOT mean N2 ↔ N3



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Shared VPC versus VPC peering

Consideration	Shared VPC	VPC Network Peering
Across organizations	No	Yes
Between VPCs in the same project	No	Yes
Network administration	Centralized	Decentralized
Basic idea	One VPC shared by multiple projects	Separate VPCs connected together

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Why Hybrid Cloud?

An on-premises database must communicate with an application in Google Cloud during migration.

Hybrid cloud = connecting on-premises infrastructure with cloud resources

Cloud VPN Securely Connects Networks over the Internet

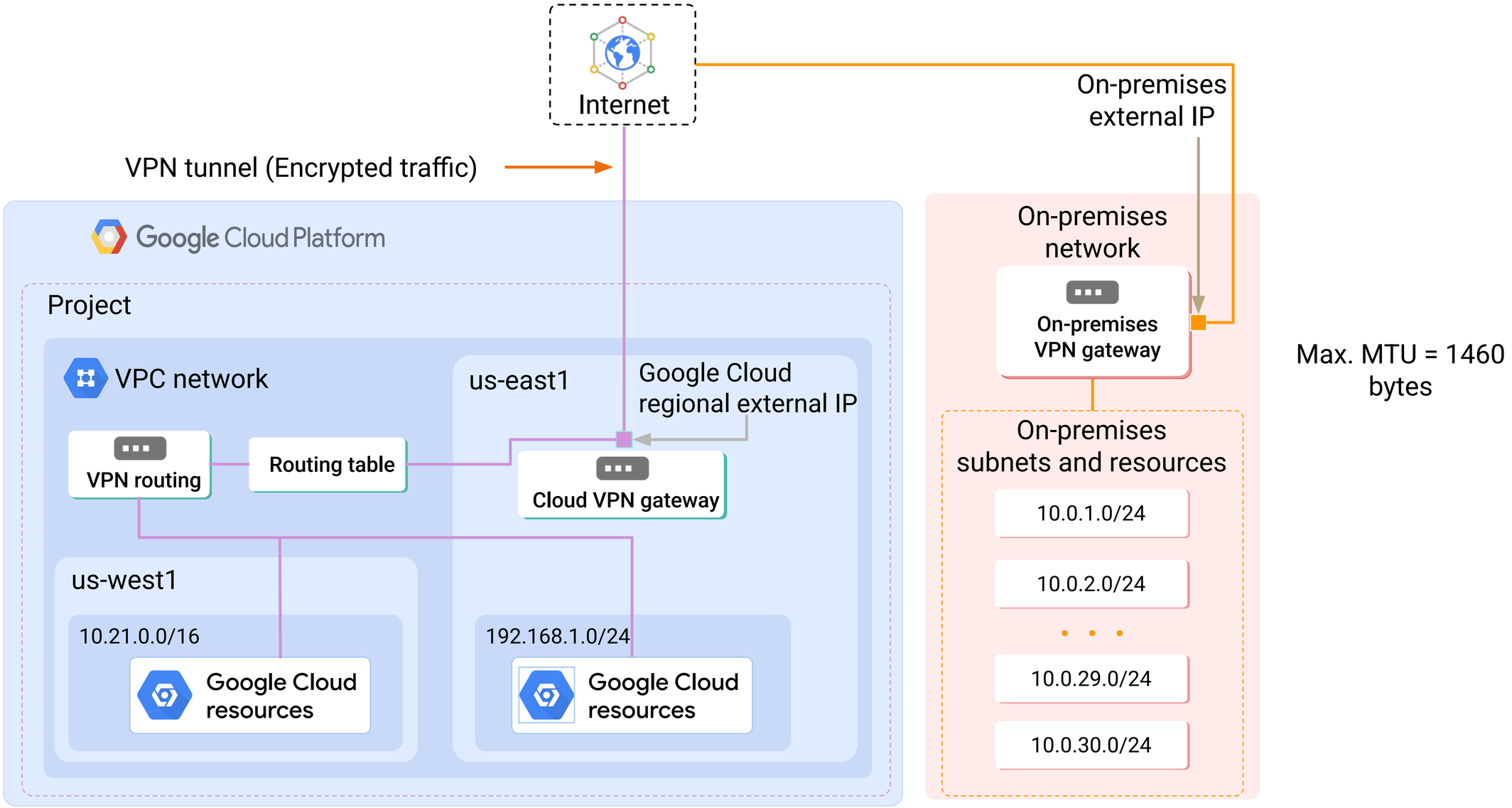
- Connects an on-premises/peer network to a Google Cloud VPC using **IPsec**
- Traffic is **encrypted** while traveling over the public Internet
- Supports **site-to-site VPN**; not client “dial-in” VPN
- **HA VPN:** dynamic routing with Cloud Router/BGP; recommended for high availability
- **Classic VPN:** static routing
- Supports IKEv1 and IKEv2



Cloud VPN = encrypted private-network connectivity over the public Internet

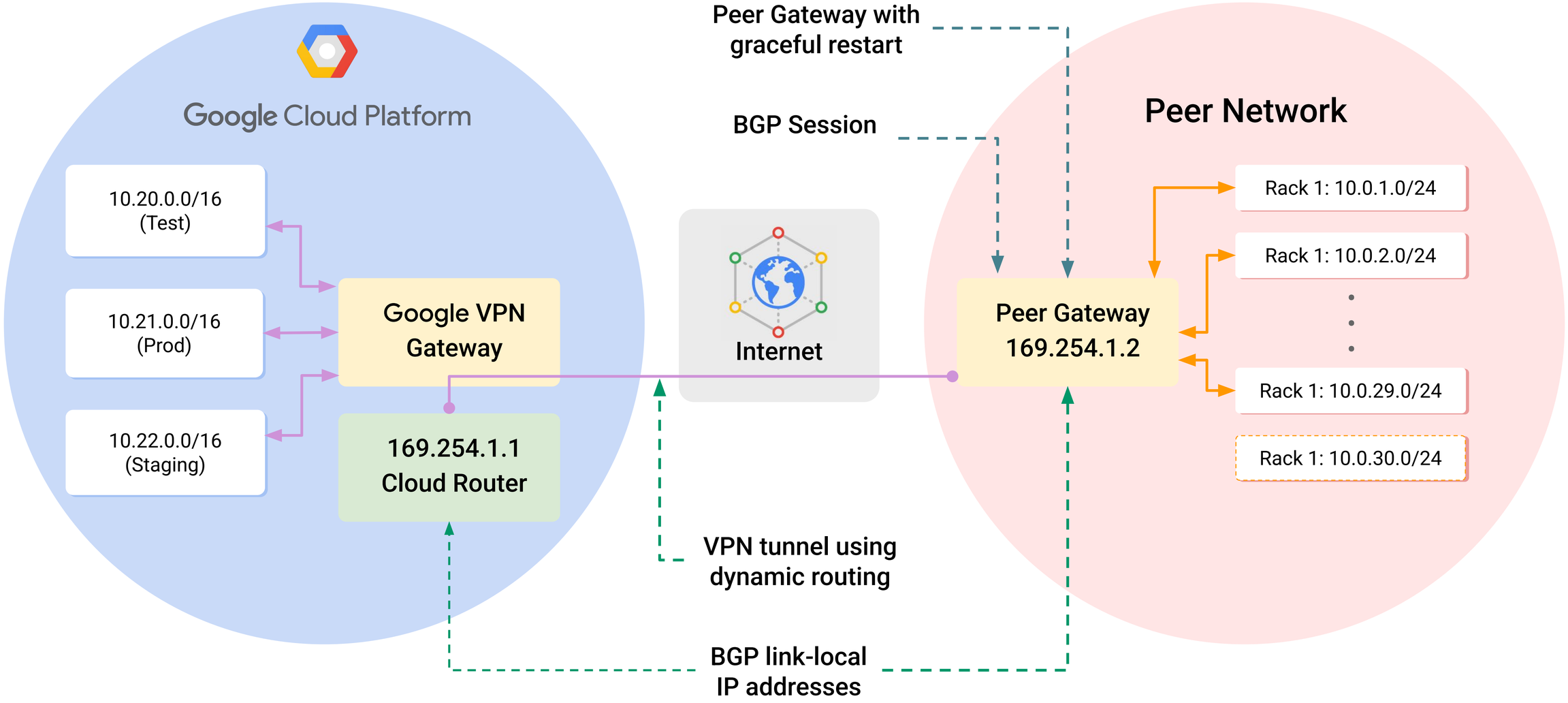
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Static VPN Topology — Conceptual Example



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HA VPN with Dynamic Routing: Cloud Router and BGP



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On-Premises Connectivity: Dedicated and Partner Interconnect.



Cloud Interconnect - Dedicated

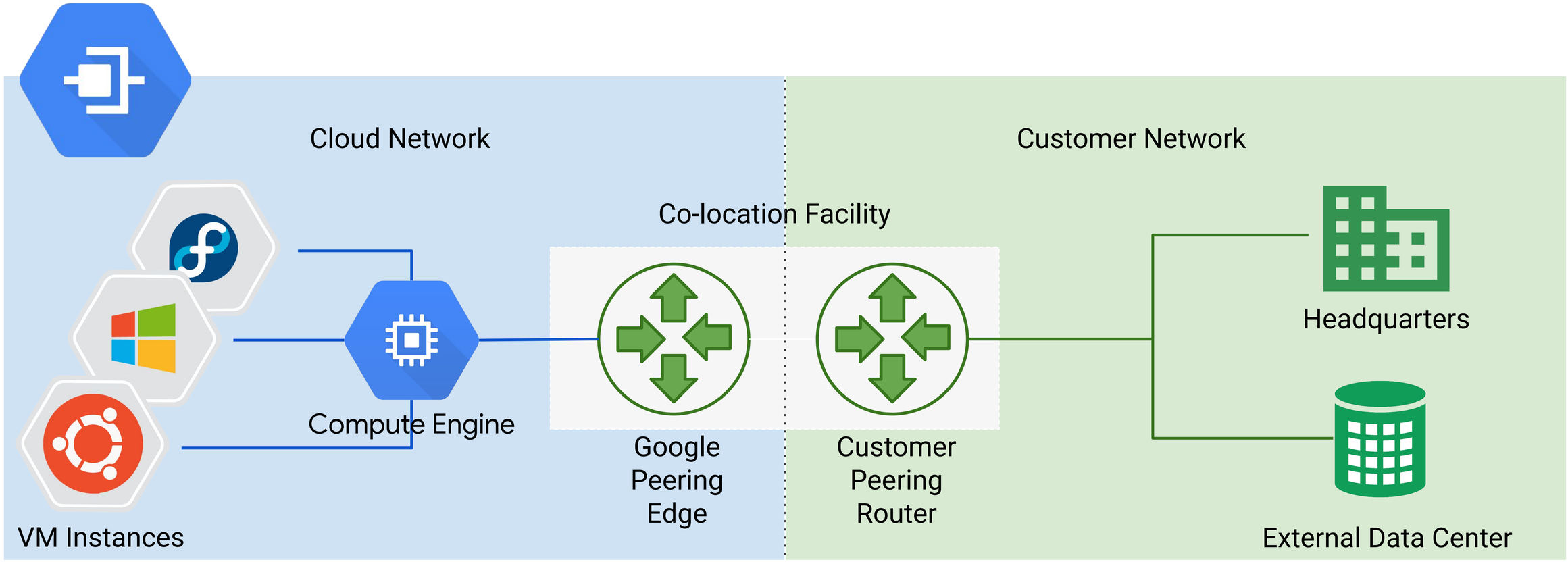


Cloud Interconnect - Partner

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Dedicated Interconnect

Dedicated Interconnect = direct physical connection from your network to Google
Best suited for organizations that need high-capacity, private connectivity and can meet Google at a supported colocation facility.

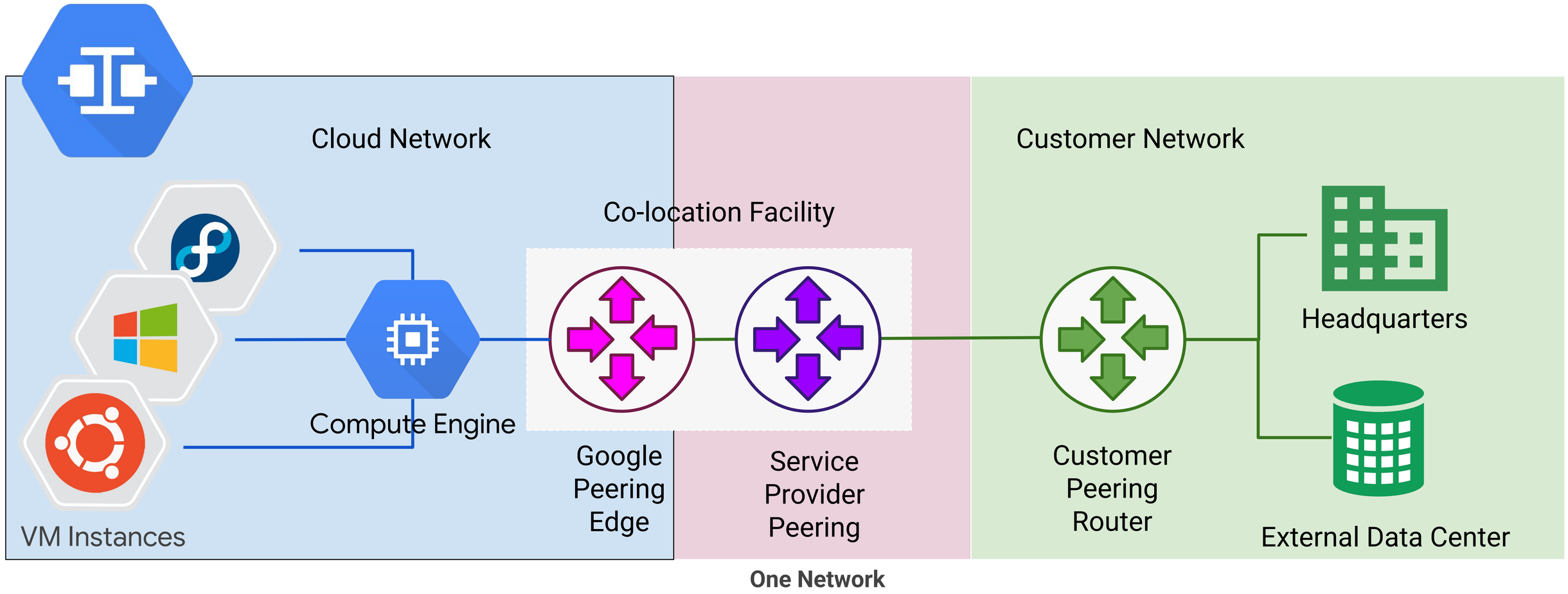


One Network

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Partner Interconnect

Partner Interconnect = private connectivity to Google through a service provider



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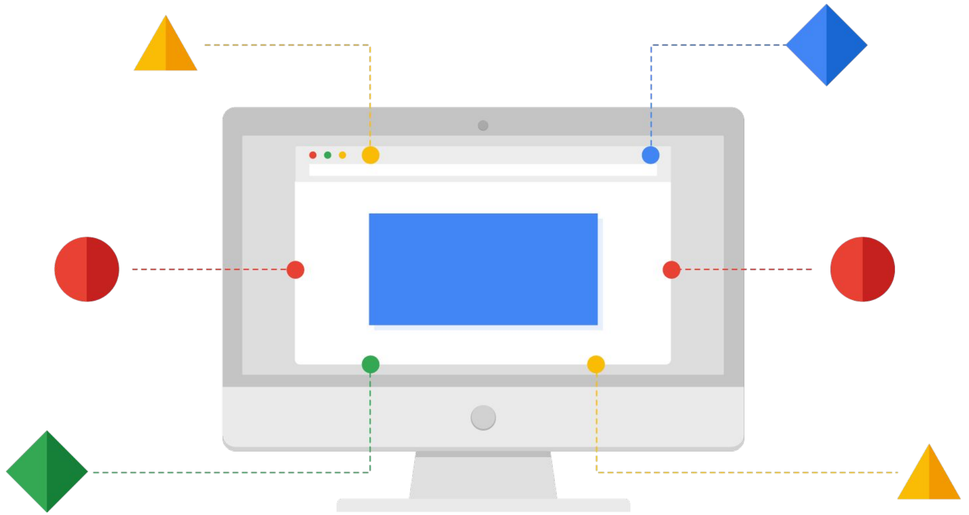
Comparing Hybrid Connectivity Options

Connection	Provides	Capacity	Requirements	Access type
IPsec VPN tunnels	Encrypted tunnel to VPC networks through the public internet	1.5 - 3.0 Gbps per tunnel	On-premises VPN gateway	Internal IP addresses
Dedicated Interconnect	Dedicated, direct connection to VPC networks	8 x 10 Gbps circuits, or 2 x 100 Gbps circuits per connection	Connection in a colocation facility	Internal IP addresses
Partner Interconnect	Dedicated bandwidth, connection to VPC network through a service provider	50 Mbps – 10 Gbps per connection	Service provider	Internal IP addresses

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Direct Peering — Direct Access to Google's Public Network

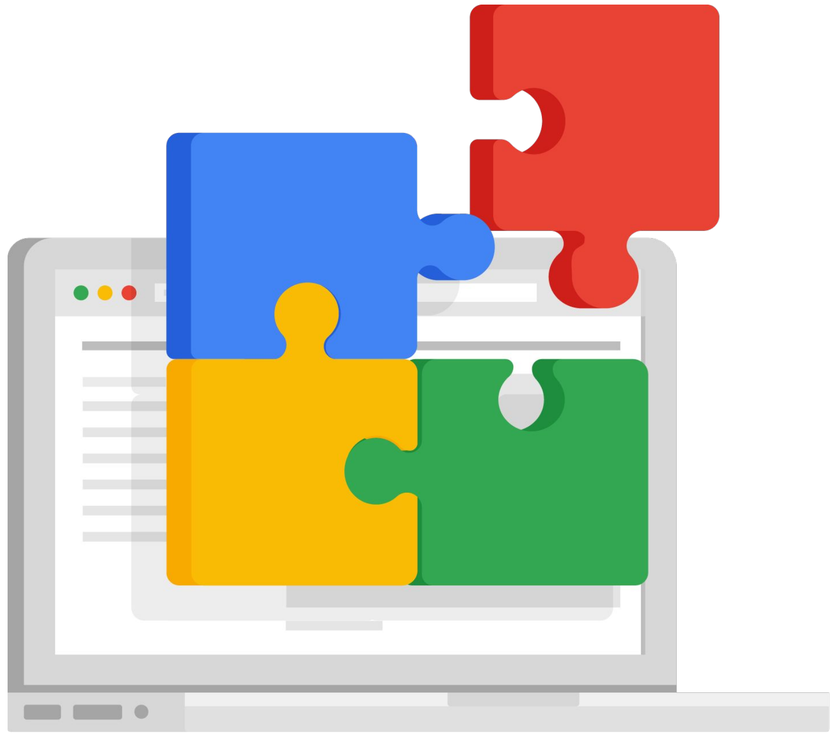
- Leverage Google’s broad-reaching edge network locations.
- Exchange BGP routes between Google and the peering entity.
- Reach all of Google’s services.
- No SLA applies.
- Peering requirements must be satisfied.



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Carrier Peering — Access Google Through a Service Provider

- An alternative if Google's peering requirements cannot be met.
- Leverage a provider's enterprise- grade network services to access Google applications.
- Get connections with higher availability and lower latency.
- No SLA offered by Google but may be offered by the provider.



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A comparison of peering options

Connection	Provides	Capacity	Requirements	Access type
Direct Peering	Dedicated, direct connection to Google's network	10 Gbps per link	Connection in Google Cloud PoPs	Public IP addresses
Carrier Peering	Peering through a service provider to Google’s public network	Varies based on partner offering	Service provider	Public IP addresses

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Summary

- **Shared VPC** lets multiple projects use a common centrally managed VPC.
- **VPC Network Peering** privately connects separate VPC networks.
- VPC Peering is **not transitive**.
- **Cloud VPN** provides encrypted connectivity over the Internet.
- **Dedicated Interconnect** provides a direct physical connection to Google.
- **Partner Interconnect** provides connectivity through a service provider.
- Choose the connection method based on network relationship, security, capacity, availability, location, and cost.

Practice: Choose the right connectivity option

1. Two separate Google Cloud projects need private communication but should remain independently administered.
2. Several projects in one organization should share a centrally managed network.
3. A small company wants an encrypted connection from its office to Google Cloud using the Internet.
4. A large enterprise needs high-capacity private connectivity directly to Google Cloud.
5. The company needs private connectivity but cannot connect directly at a Google colocation facility.

